Aparajita Kashyap, MA, MPhil

Biomedical Informatics
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Education

2022– **PhD, Biomedical Informatics.** Columbia University. New York, NY. *Expected August 2027*
MPhil, Biomedical Informatics. May 2026.
MA, Biomedical Informatics. January 2024.
Advisors: Drs. Shalmali Joshi and Noémie Elhadad

2018–22 **BA, Biophysics.** Johns Hopkins University. Baltimore, MD.
Minors: Applied Mathematics and Statistics, Space Science and Technology

Work Experience

2026– **AI/ML Graduate Research Intern.** Optum AI Science.

2022– **Graduate Research Assistant.** Department of Biomedical Informatics, Columbia University.

2020-2022 **Undergraduate Research Assistant.** Department of Biophysics, Johns Hopkins University.

2020–21 **CIRCUIT Intern.** Johns Hopkins University Applied Physics Laboratory.

Publications

Publications - Conference and Journal

1. **A Kashyap**, S Matijevic, N Elhadad, SA Kushner, and S Joshi. A pipeline for enabling path-specific causal fairness in observational health data. *ACM Conference on Fairness, Accountability, and Transparency*, 2026. **Best Paper Award (3 of 325)**
2. **A Kashyap**, N Elhadad, SA Kushner, and S Joshi. Schizophrenia risk prediction following antecedent psychosis using machine learning. *Schizophrenia*, 2026. *Accepted for publication.*
3. The Justice Informatics Collaborative: **A Kashyap***, CJ Allsman*, EA Campbell, PM Desai, SG Volpe, BP Massey, TJ Bright, S Bakken, OJ Bear Don't Walk IV, and A Pichon. Contextualizing key principles to promote a justice-oriented informatics research agenda: Proceedings and reflections from an amia workshop. *Journal of the American Medical Informatics Association (JAMIA)*, 2026
4. Z Jing, V Jeanselme, Y Kobayashi, SA Lee, C Pang, **A Kashyap**, Y Li, X Jiang, and S Joshi. One loss to rule them all: Marked time-to-event for structured ehr foundation models. *Advances in Neural Information Processing Systems*, 2026. **Previously spotlighted in SD4H ICML Workshop 2026.**
5. **A Kashyap**, M Aziz, TY Sun, S Lipsky-Gorman, J Opoku-Anane, and N Elhadad. Investigating racial disparities in drug prescriptions for patients with endometriosis. *npj women's health*, 2025
6. **A Kashyap**, W Wang, and BA Camley. Tradeoffs in concentration sensing in dynamic environments. *Biophysical Journal*, 2024
7. HP Cowley, MS Robinette, JK Matelsky, D Xenes, **A Kashyap**, NF Ibrahim, ML Robinson ML, S Zeger, BT Garibaldi, and W Gray-Roncal. Using machine learning on clinical data to identify unexpected patterns in groups of covid-19 patients. *Scientific Reports*, 13, 2023

* Contributed equally, co-first authors

Preprints

1. Jeanselme V*, Jing Z*, **Kashyap A***, Pang C*, Pollet F*, Choi YS, Jiang X, Kobayashi Y, Li Y, Matijevic S, Natarajan K, and Joshi S. Fomoh: A clinically meaningful foundation model evaluation for structured electronic health records. arXiv [preprint], 2025
2. E Alsentzer, ML Charpignon, B Chen... **A Kashyap**..., H Jeong, and K Thakoor. Reflections from research roundtables at the conference on health, inference, and learning (chil) 2025. arXiv [preprint], 2025
3. H Reyes Nieva, **A Kashyap**, EA Voss, A Ostropolets, A Anand, M Ketenci, FJ DeFalco, Natarajan K, YS Choi, Y Li, MN Allen, SA Guang, and N Elhadad. The impact of evolving endometriosis guidelines on diagnosis and observational health studies. medRxiv [preprint], 2024

Peer-Reviewed Abstracts

1. **A Kashyap**, N Elhadad, SA Kushner, and S Joshi. Population-level characterization of individuals with whole genome sequencing samples in the all of us research program. *OHDSI Global Symposium*, 2026
2. **A Kashyap (speaker)**, S Kushner, N Elhadad, and S Joshi. Leveraging data-driven analyses to improve subpopulation robustness of a schizophrenia prediction model. *AMIA Annual Symposium*, 2025. Oral Presentation
3. PM Desai, A Raj, D Albers, **A Kashyap**, and L Mamykina. Something similar: Exploring the usefulness of an on-the-spot meal recommendation system for health goal attainment. *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025
4. H Reyes Nieva, **A Kashyap**, EA Voss, A Ostropolets, A Anand, M Ketenci, S Guang, MN Allen, FJ DeFalco, YS Choi, Y Li, K Natarajan, P Ryan, and N Elhadad. Expanding endometriosis diagnostic guidelines captures more women with non-classical symptoms, different care-seeking patterns, and lack of access to surgical care. *Endometriosis: A Look at Current Practices and Emerging Science, New York Academy of Sciences*, 2025. Poster
5. **A Kashyap**, N Elhadad, SA Kushner, and S Joshi. Leveraging machine learning explanations to assess the appropriateness of using demographic information during schizophrenia onset prediction. *AMIA Annual Symposium*, 2024. Poster
6. **A Kashyap***, C Allsman*, E Campbell, P Desai, S Volpe, B Massey, S Bakken, OJ IV Bear Don't Walk, and A Pichon. A broad perspective on justice-oriented research in biomedical informatics. *AMIA Annual Symposium*, 2024. Oral Presentation
7. P Nong, S El-Azab, **A Kashyap (speaker)**, D Newman-Griffis, and OJ IV Bear Don't Walk. Making justice and equity tangible in informatics. *AMIA Annual Symposium*, 2024. Panelist
8. H Reyes Nieva, **A Kashyap**, EA Voss, A Ostropolets, A Anand, M Ketenci, FJ DeFalco, Natarajan K, YS Choi, Y Li, MN Allen, SA Guang, and N Elhadad. The impact of evolving endometriosis guidelines on diagnosis and observational health studies. *OHDSI Global Symposium*, 2024
9. A Kolo, C Pang, E Choi... **A Kashyap**..., N Oufattole, T Bergamaschi, and X Jiang. Meds decentralized, extensible validation (meds-dev) benchmark: Establishing reproducibility and comparability in ml for health. *Machine Learning For Health (ML4H) Demo Track*, 2024
10. A Pichon, S Volpe, **A Kashyap (speaker)**, P Desai, T Anand, E Campbell, K Schiffer, C Diamond, B Massey, C Allsman, A Newbury, L Richter, T Bright, S Bakken, and OJ IV Bear Don't Walk. Synthesizing gaps and priorities for a justice informatics research agenda. *AMIA Annual Symposium*, 2023. Oral Presentation
11. **A Kashyap (speaker)**, M Aziz, TY Sun, S Lipsky-Gorman, J Opoku-Anane, and N Elhadad. Investigating racial inequality in drug prescriptions for patients with endometriosis. *World Congress on Endometriosis*, 2023. Oral Presentation

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12. **A Kashyap**, M Aziz, TY Sun, S Lipsky-Gorman, J Opoku-Anane, and N Elhadad. Investigating racial inequality in drug prescriptions for patients with endometriosis. *National Library of Medicine T15 Training Conference, Stanford University*, 2023. Poster
13. **A Kashyap**, TY Sun, and N Elhadad. Data-driven identification of symptoms of interest in the early detection of endometriosis. *World Congress on Endometriosis*, 2023. Poster
14. S Tang, V Chandra, **A Kashyap**, W Kilburn, C Spencer, K Yaovatsakul, J Nguyen, D Roberts, and M Shelhamer. Multivariate analysis of human health and performance in an acute spaceflight simulation. *NASA Human Research Program Investigator's Workshop*, 2020. Poster
15. S Tang, M Tan, V Chandra, **A Kashyap**, W Kilburn, D Roberts, and M Shelhamer. Vestibular and postural assessment device and methods. *NASA Human Research Program Investigator's Workshop*, 2020. Poster

Invited Talks and Lectures

1. A pipeline for enabling path-specific causal fairness in observational health data. *Fairness of AI in Medical Imaging Symposium (SIG-FAIMI)*, 2026
2. Machine learning for enabling fair precision psychiatry. *SNF Center for Precision Psychiatry and Mental Health*, 2026
3. Leveraging data-driven analyses to characterize subpopulation robustness. *University of Washington Biomedical Informatics Research Colloquium*, 2025
4. Machine learning predicts diagnostic transition to schizophrenia using administrative claims data. *SNF Center for Precision Psychiatry and Mental Health*, 2025
5. Machine learning for time series data. *Machine Learning for Healthcare, Columbia University*, 2024

Professional Engagement and Leadership

2025– **Vagelos Computational Science Center, Barnard College.** Graduate Assistant for Public Interest Technology. *Developing extracurricular programming and providing mentorship around Public Interest Technology for undergraduate students at Barnard College. Programming includes organization two submissions to the 2025 NASA Space Apps Challenge and participation in New York City Open Data Week (data analysis, interviews, short documentary).*

2022–25 **Justice Informatics Collaborative.** Co-leader. *The group engages trainees and the broader community of informaticians to advance justice-oriented science. Group activities include hosting a workshop and authoring manuscripts; department activities include facilitating discussions around seminars; informatics community activities include organizing a workshop at the AMIA Annual Symposium and synthesizing insights to present at AMIA meetings.*

2023–24 **Disability Ethics, Intersectionality & AI/ML Bias Speaker Series.** Series Coordinator. *Organized a speaker series and led discussion groups relating to disability ethics within informatics and AI/ML.*

2023–24 **Columbia Precision Medicine & Society.** Graduate fellow. *Organized a speaker series and led discussion groups relating to precision medicine, ethics, privacy, and fairness.*

Teaching Experience

Courses

2024 **Machine Learning in Healthcare** Teaching Assistant

2023 **Acculturation to Programming and Statistics** Teaching Assistant, Lecturer for Probability and Statistics Unit